

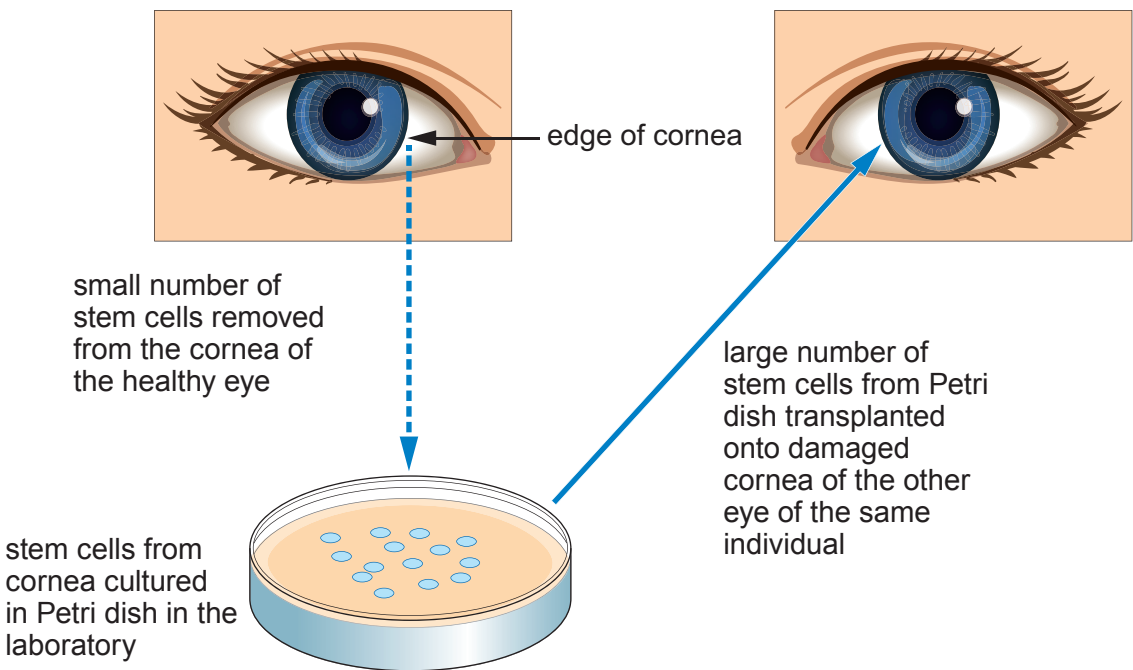
9. The eye is a complex organ made up of many tissues. Diseases of the eye occur when one or more of these tissues are damaged or stop functioning. Stem cell treatments are now being developed and used to treat many eye diseases.

Example 1 – Repairing the cornea (This treatment is now licensed for use.)

Stem cells at the edge of the cornea replace cells that are constantly being worn away. These stem cells may also be damaged by burns, radiation, genetic disorders, infection and drug use.

The diagram below shows a stem cell treatment to repair the cornea. This treatment uses stem cells from a healthy part of the cornea of an affected individual.

Diagram 1

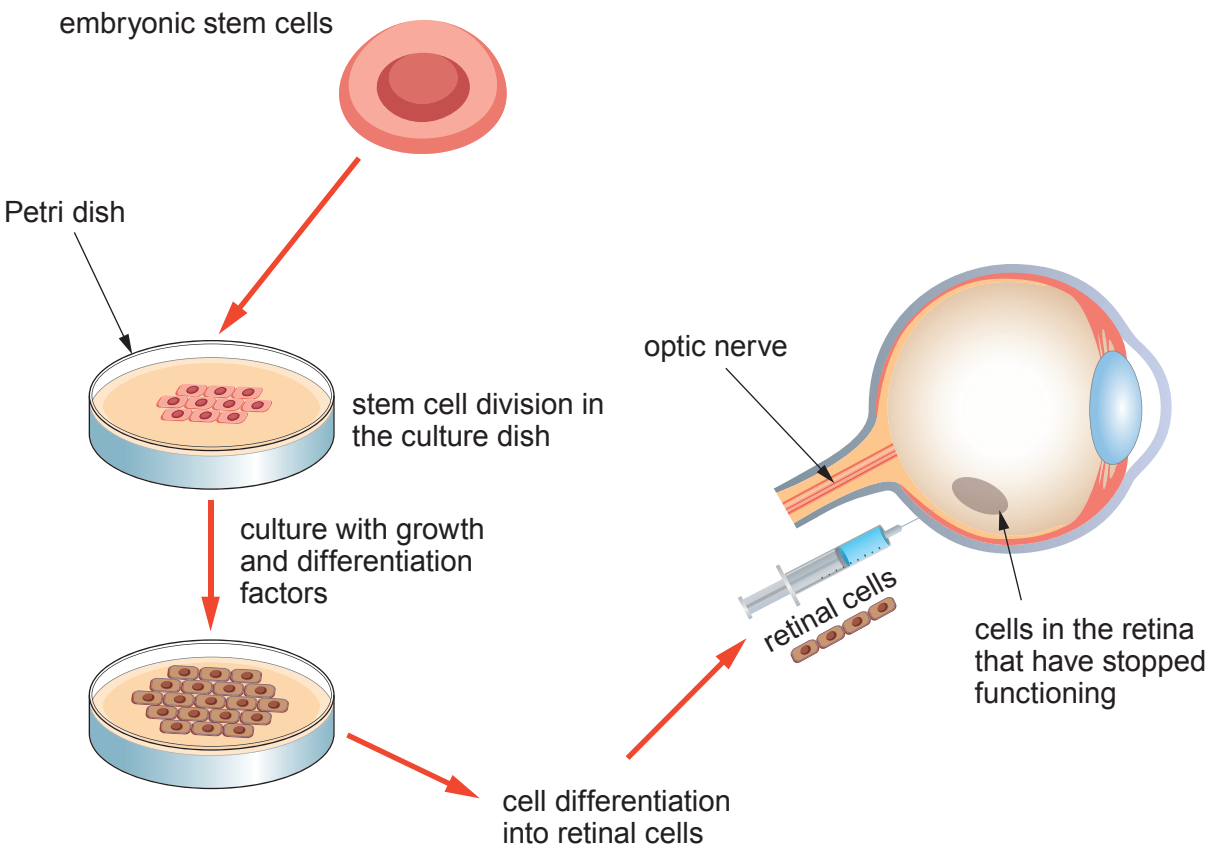


Example 2 – Repairing the Retina (This treatment is currently being researched and tested.)

In age-related macular degeneration (ARMD), cells in the retina stop functioning.

The diagram below shows a stem cell treatment for ARMD that uses embryonic stem cells to replace non-functioning cells in the retina.

Diagram 2



(a) State the function of the cornea. [1]

.....

.....

(b) Explain the meaning of the term *stem cell*. [1]

.....

.....



Examiner
only

(c) State the name of the type of cell division that occurs in the Petri dishes in diagrams **1** and **2** and explain its significance in the treatments. [3]

.....

.....

.....

(d) Suggest why example **1** does not work if both eyes are badly damaged. [1]

.....

.....

(e) State **two** advantages of using stem cells as a treatment in example **1** compared to their use in example **2**. [2]

.....

.....

.....

.....

8

END OF PAPER

